DEPI
Digital Egypt Pioneers Initiative
Track: **Professional Data Analyst**

Sales Performance Analysis

Bike and accessories sales data turned into clear answers on what sells, who buys, and where.

- 60K+ order lines
- 18K+ customers
- 295 products

Business Problem

Sales are spread across many products, categories, countries and customer types. Without a clear view, it is hard to see what drives revenue and margin, or where to focus.

Proposed Solution

A star-schema model linking sales, products and customers, with KPIs, segmentation and a sales forecast, delivered as an interactive Power BI dashboard.

Project Roadmap

SQL · Python · Power BI · DAX

WEEK 1 — Data Model & Cleaning
- Star schema: sales fact + product & customer dims
- Fix types, dates, keys and duplicates
- Derive revenue, margin, age & ship time
- Data dictionary & quality report

WEEK 2 — Analysis & KPIs
- Revenue, orders, AOV & margin KPIs
- Trends by month, category, country
- Top & bottom products and customers
- Segments by age, gender, marital status

WEEK 3 — Sales Forecasting
- Forecast monthly sales
- Compare models, validate accuracy
- Seasonality & category outlook
- Actual vs predicted, assumptions

WEEK 4 — Dashboard & Report
- Interactive Power BI + DAX model
- KPIs, trends, segments, forecast
- Management decision view
- Final report & presentation

Technology Stack

SQL Modeling & cleaning	**Python** Prep & analysis	**pandas** Data wrangling
NumPy Calculations	**Matplotlib** Charts	**scikit-learn** Forecast models
Power BI Dashboard	**DAX** KPI measures	**Excel** Quick checks

Project Outcomes

- Interactive dashboard
- Sales forecast
- Star-schema data model
- Report & recommendations

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